8.4 Hydraulic Turbines

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Introduction

A hydraulic turbine is a mechanical device that converts the potential energy associated with a difference in water elevation **(head)** into useful work. Modern hydraulic turbines are the result of many years of gradual development. Economic incentives have resulted in the development of very large units (exceeding 800 mW in capacity) with efficiencies that are sometimes in excess of 95%.

The emphasis on the design and manufacture of very large turbines is shifting to the production of smaller units, especially in developed nations, where much of the potential for developing large-baseload plants has been realized. At the same time, the escalation in the cost of energy has made many smaller sites economically feasible and has greatly expanded the market for smaller turbines. The increased value of energy also justifies the cost of refurbishment and increasing the capacity of older facilities. Thus, a new market area is developing for updating older turbines with modern replacement **runners** having higher efficiency and greater capacity.

General Description

Typical Hydropower Installation

As shown schematically in Figure 8.4.1, the hydraulic components of a hydropower installation consist of an intake, penstock, guide vanes or distributor, turbine, and **draft tube**. Trash racks are commonly provided to prevent ingestion of debris into the turbine. Intakes usually require some type of shape transition to match the passageway to the turbine and also incorporate a gate or some other means of stopping the flow in case of an emergency or for turbine maintenance. Some types of turbines are set in an open flume; others are attached to a closed-conduit penstock.

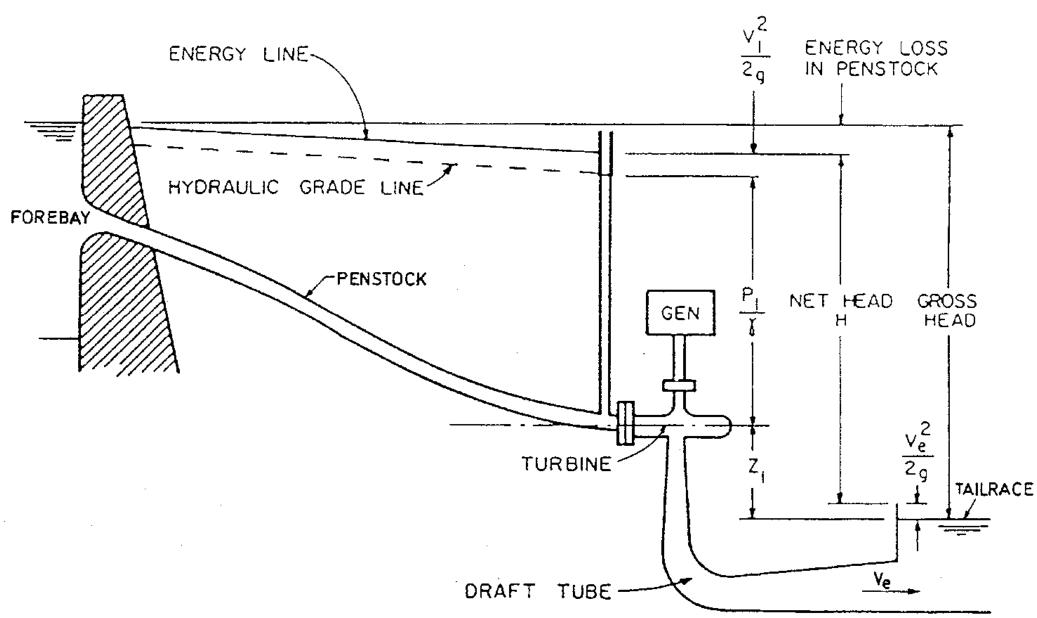

FIGURE 8.4.1 Schematic of a hydropower installation.

Turbine Classification

There are two types of turbines, denoted as impulse and reaction. In an impulse turbine, the available head is converted to kinetic energy before entering the **runner**; the power available is extracted from the flow at approximately atmospheric pressure. In a reaction turbine, the runner is completely submerged and both the pressure and the velocity decrease from inlet to outlet. The velocity head in the inlet to the turbine runner is typically less than 50% of the total head available.

Impulse Turbines. Modern impulse units are generally of the Pelton type and are restricted to relatively high-head applications (Figure 8.4.2). One or more jets of water impinge on a wheel containing many curved buckets. The jet stream is directed inward, sideways, and outward, thereby producing a force on the bucket, which in turn results in a torque on the shaft. All kinetic energy leaving the runner is "lost." A draft tube is generally not used since the runner operates under approximately atmospheric pressure and the head represented by the elevation of the unit above tailwater cannot be utilized.* Since this is a high-head device, this loss in available head is relatively unimportant. As will be shown later, the Pelton wheel is a low-**specific-speed** device. Specific speed can be increased by the addition of extra nozzles, the specific speed increasing by the square root of the number of nozzles. Specific speed can also be increased by a change in the manner of inflow and outflow. Special designs such as the Turgo or cross-flow turbines are examples of relatively high specific speed impulse units (Arndt, 1991).

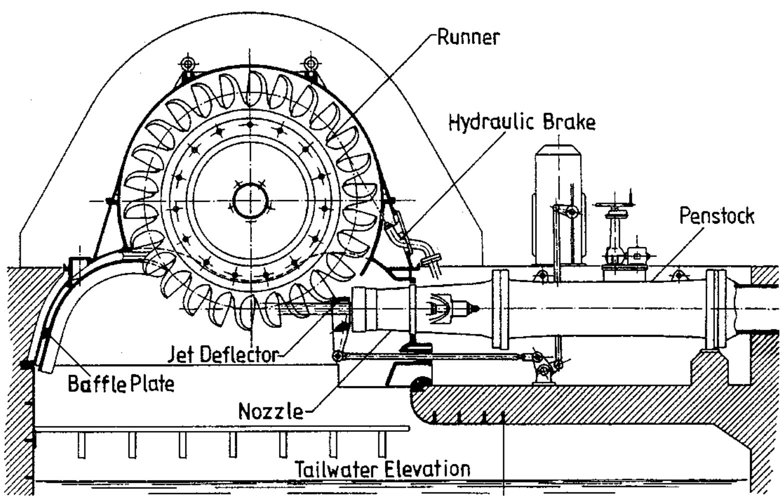

FIGURE 8.4.2 Cross section of a single wheel, single jet Pelton turbine. This is the third-highest-head pelton turbine in the world, H = 1447 m, n = 500 rpm, P = 35.2 MW, $N_s \sim 0.038$. (Courtesy of Vevey Charmilles Engineering Works, Adapted from J. Raabe, *Hydro Power: The Design, Use, and Function of Hydromechanical, Hydraulic, and Electrical Equipment*, VDI Verlag, Dusseldorf, Germany.)

Most Pelton wheels are mounted on a horizontal axis, although newer vertical-axis units have been developed. Because of physical constraints on orderly outflow from the unit, the maximum number of nozzles is generally limited to six or fewer. While the power of a reaction turbine is controlled by the **wicket gates**, the power of the Pelton wheel is controlled by varying the nozzle discharge by means of an automatically adjusted needle, as illustrated in Figure 8.4.2. Jet deflectors, or auxiliary nozzles are provided for emergency unloading of the wheel. Additional power can be obtained by connecting two wheels to a single generator or by using multiple nozzles. Since the needle valve can throttle the flow while maintaining essentially constant jet velocity, the relative velocities at entrance and exit remain unchanged, producing nearly constant efficiency over a wide range of power output.

* In principle, a draft tube could be used, which requires the runner to operate in air under reduced pressure. Attempts at operating an impulse turbine with a draft tube have not met with much success.

Reaction Turbines. Reaction turbines are classified according to the variation in flow direction through the runner. In radial- and mixed-flow runners, the flow exits at a radius different than from the radius at the inlet. If the flow enters the runner with only radial and tangential components, it is a radial-flow machine. The flow enters a mixed-flow runner with both radial and axial components. Francis turbines are of the radial- and mixed-flow types, depending on the design specific speed. A Francis turbine is illustrated in Figure 8.4.3.

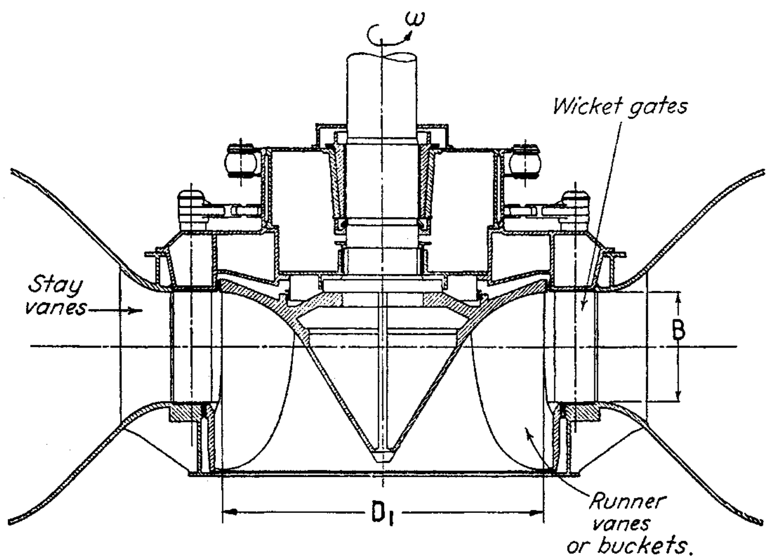

FIGURE 8.4.3 Francis turbine, N_s ~ 0.66. (Adapted from J.W. Daily, in *Engineering Hydraulics*, H. Rouse, Ed., New York, 1950. With permission.)

Axial-flow propeller turbines are generally either of the fixed-blade or Kaplan (adjustable-blade) variety. The "classical" propeller turbine, illustrated in Figure 8.4.4, is a vertical-axis machine with a scroll case and a radial wicket gate configuration that is very similar to the flow inlet for a Francis turbine. The flow enters radially inward and makes a right-angle turn before entering the runner in an axial direction. The Kaplan turbine has both adjustable runner blades and adjustable wicket gates. The control system is designed so that the variation in blade angle is coupled with the wicket gate setting in a manner which achieves best overall efficiency over a wide range of flow rates.

Some modern designs take full advantage of the axial-flow runner; these include the tube, bulb, and Straflo types illustrated in Figure 8.4.5. The flow enters and exits the turbine with minor changes in direction. A wide variation in civil works design is also permissible. The tubular-type can be fixed propeller, semi-Kaplan, or fully adjustable. An externally mounted generator is driven by a shaft which extends through the flow passage either upstream or downstream of the runner. The bulb turbine was originally designed as a high-output, low-head unit. In large units, the generator is housed within the bulb and is driven by a variable-pitch propeller at the trailing end of the bulb. Pit turbines are similar in principle to bulb turbines, except that the generator is not enclosed in a fully submerged compartment (the bulb). Instead, the generator is in a compartment that extends above water level. This improves access to the generator for maintenance.

Principles of Operation

Power Available, Efficiency

The power that can be developed by a turbine is a function of both the head and flow available:

$$P = \eta \rho g Q H \tag{8.4.1}$$

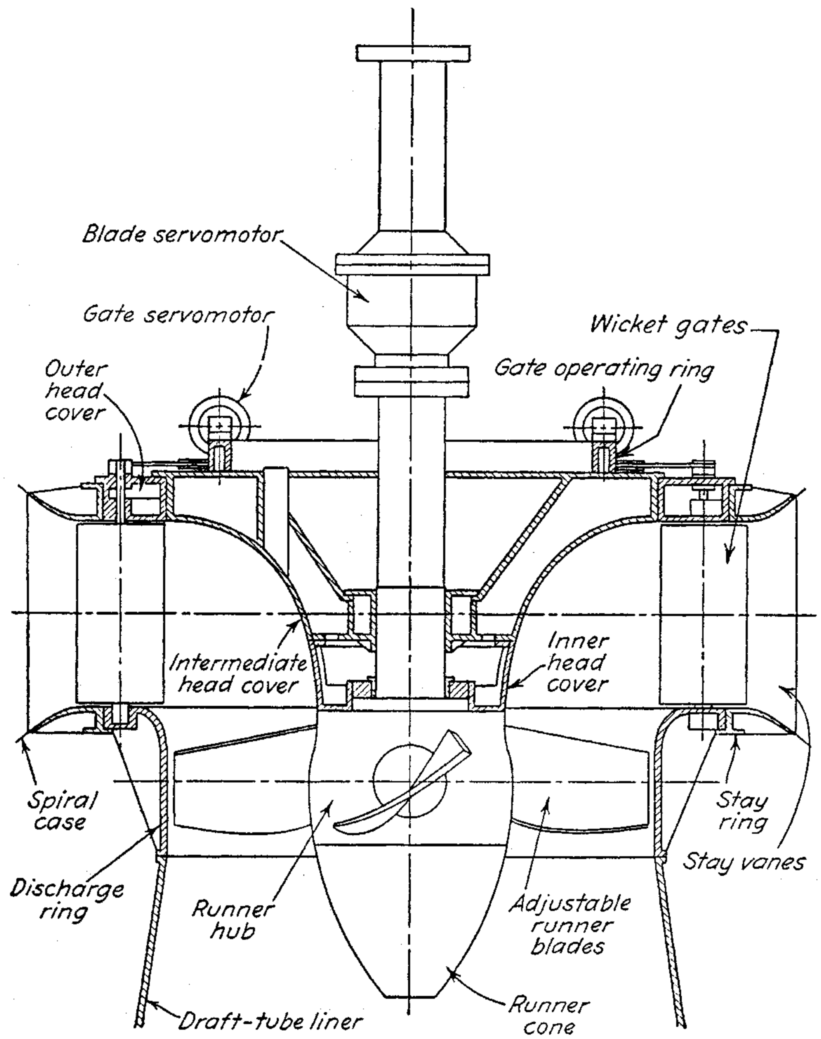

FIGURE 8.4.4 Smith-Kaplan axial flow turbine with adjustable-pitch runner blades $N_s \sim 2.0$. (From J.W. Daily, in *Engineering Hydraulics*, H. Rouse, Ed., New York, 1950. With permission.)

where η is the turbine efficiency, ρ is the density off water (kg/m^3), g is the acceleration due to gravity (m/sec^2), Q is the flow rate (m^3/sec), and H is the *net head* in meters. Net head is defined as the difference between the total head at the inlet to the turbine and the total head at the tailrace as illustrated in Figure 8.4.1. Different definitions of net head are used in practice, which depend on the value of the exit velocity head, $V_e^2/2g$, used in the calculation. The International Electrotechnical Test Code uses the velocity head at the draft tube exit.

The efficiency depends on the actual head and flow utilized by the turbine runner, flow losses in the draft tube, and the frictional resistance of mechanical components.

Similitude and Scaling Formulae

Under a given head, a turbine can operate at various combinations of speed and flow depending on the inlet settings. For reaction turbines, the flow into the turbine is controlled by the wicket gate angle, α. The flow is typically controlled by the nozzle opening in impulse units. Turbine performance can be described in terms of nondimensional variables:

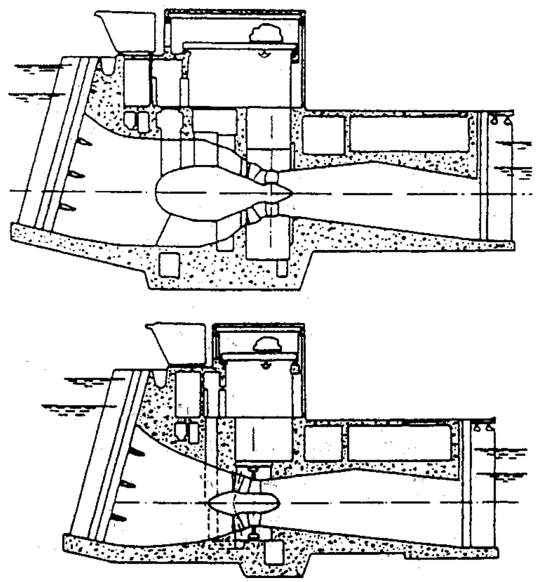

FIGURE 8.4.5 Comparison between bulb (upper) and Straflow (lower) turbines. (Courtesy of U.S. Department of Energy.)

$$\psi = \frac{2gH}{\omega^2 D^2} \tag{8.4.2}$$

$$\phi = \frac{Q}{\sqrt{2gH}D^2} \tag{8.4.3}$$

where ω is the rotational speed of the turbine in radians per second and D is the diameter of the turbine.

The hydraulic efficiency of the runner alone is given by

$$\eta_h = \frac{\phi}{\sqrt{\psi}}\left(C_1 \cos\alpha_1 - C_2 \cos\alpha_2\right) \tag{8.4.4}$$

where C_1 and C_2 are constants that depend on the specific turbine configuration and α_1 and α_2 are the inlet and outlet angles that the absolute velocity vectors make with the tangential direction. The value of $\cos\alpha_2$ is approximately zero at peak efficiency. The terms ϕ, ψ, α_1, and α_2 are interrelated. By using model test data, isocontours of efficiency can be mapped in the ϕ-ψ plane. This is typically referred to as a hill diagram as shown in Figure 8.4.6.

The specific speed is defined as

$$N_s \equiv \frac{\omega\sqrt{Q}}{\left(2gH\right)^{3/4}} = \sqrt{\frac{\phi}{\psi}} \tag{8.4.5}$$

A given specific speed describes a specific combination of operating conditions that ensures similar flow patterns and same efficiency in geometrically similar machines regardless of size and rotational speed of the machine. It is customary to define the design specific speed in terms of the value at the design head and flow where peak efficiency occurs. The value of specific speed so defined permits a classification of different turbine types.

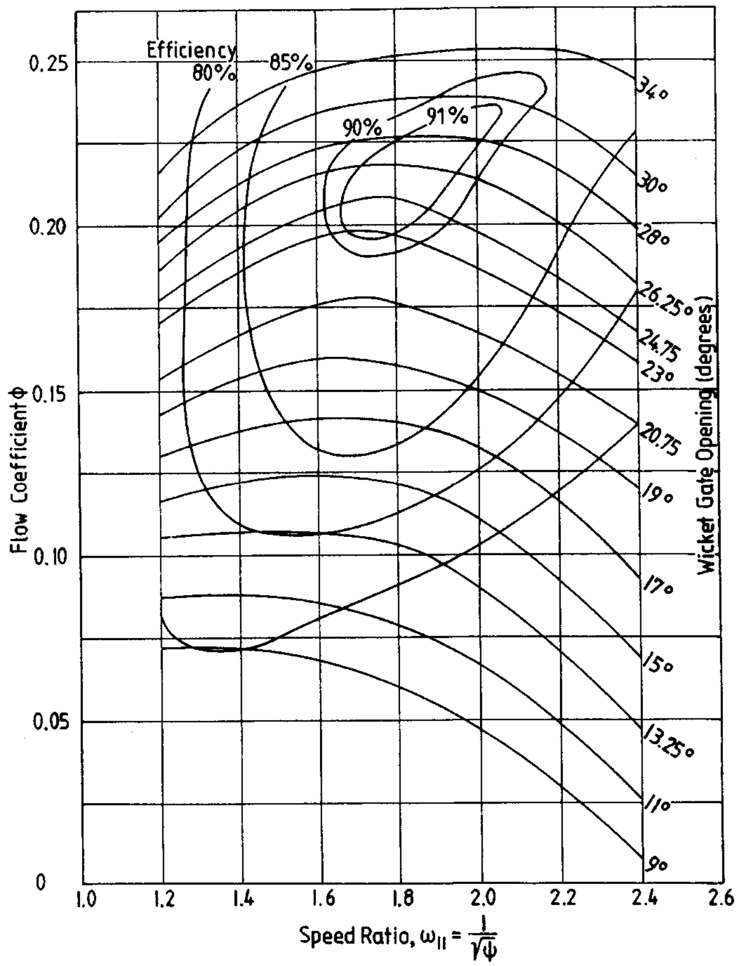

FIGURE 8.4.6 Typical hill diagram. (Adapted from T.L. Wall, Draft tube surging times two: the twin vortex problem, in *Hydro Rev.* 13(1): 60–69, 1994. With permission.)

The specific speed defined herein is dimensionless. Many other forms of specific speed exist which are dimensional and have different numerical values depending on the system of units used (Arndt, 1991).* The similarity arguments used to arrive at the concept of specific speed indicate that a given machine of diameter D operating under a head H will discharge a flow Q and produce a torque T and power P at a rotational speed ω given by

$$Q = \phi D^2 \sqrt{2gH} \tag{8.4.6}$$

$$T = T_{11} \rho D^3 2gH \tag{8.4.7}$$

$$P = P_{11} \rho D^2 (2gH)^{3/2} \tag{8.4.8}$$

$$\omega = \frac{2u_1}{D} = \omega_{11} = \frac{\sqrt{2gH}}{D} \qquad \left[\omega_{11} = \frac{1}{\sqrt{\psi}}\right] \tag{8.4.9}$$

* The literature also contains two other minor variations of the dimensionless form. One differs by a factor of $1/\pi^{1/2}$ and the other by $2^{3/4}$.

with

$$P_{11} = T_{11}\omega_{11} \tag{8.4.10}$$

where T_{11}, P_{11}, and ω_{11} are also nondimensional.* In theory, these coefficients are fixed for a machine operating at a fixed value of specific speed, independent of the size of the machine. Equations 8.4.6 through 8.4.10 can be used to predict the performance of a large machine using the measured characteristics of a smaller machine or model.

Factors Involved in Selecting a Turbine

Performance Characteristics

Impulse and reaction turbines are the two basic types of turbines. They tend to operate at peak efficiency over different ranges of specific speed. This is due to geometric and operational differences.

Impulse Turbines. Of the head available at the nozzle inlet, a small portion is lost to friction in the nozzle and to friction on the buckets. The rest is available to drive the wheel. The actual utilization of this head depends on the velocity head of the flow leaving the turbine and the setting above tailwater. Optimum conditions, corresponding to maximum utilization of the head available, dictate that the flow leaves at essentially zero velocity. Under ideal conditions, this occurs when the peripheral speed of the wheel is one half the jet velocity. In practice, optimum power occurs at a speed coefficient, ω_{11}, somewhat less than 1.0. This is illustrated in Figure 8.4.7. Since the maximum efficiency occurs at fixed speed for fixed H, V_j must remain constant under varying flow conditions. Thus, the flow rate Q is regulated with an adjustable nozzle. However, maximum efficiency occurs at slightly lower values of ω_{11} under partial power settings. Present nozzle technology is such that the discharge can be regulated over a wide range at high efficiency.

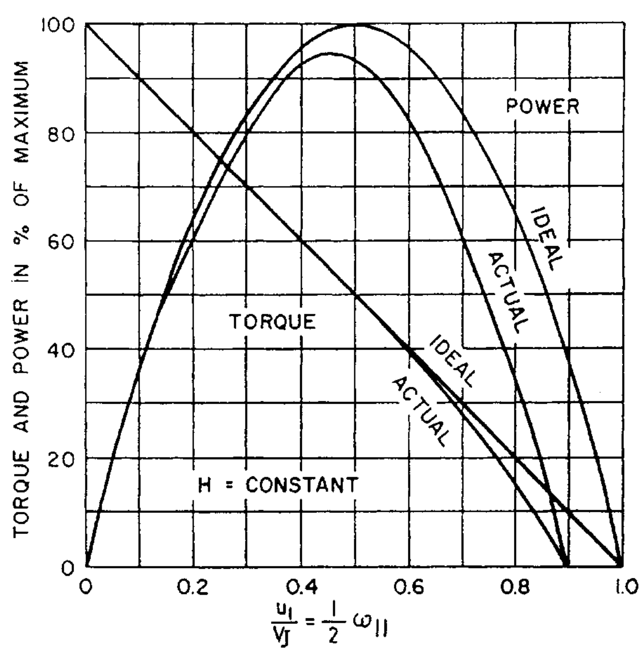

FIGURE 8.4.7 Ideal and actual variable-speed performance for an impulse turbine. (Adapted from J.W. Daily, in *Engineering Hydraulics*, H. Rouse, Ed., New York, 1950. With permission.)

* The reader is cautioned that many texts, especially in the American literature, contain dimensional forms of T_{11}, P_{11}, and ω_{11}.

A given head and penstock configuration establishes the optimum jet velocity and diameter. The size of the wheel determines the speed of the machine. The design specific speed is approximately

$$N_s = 0.77\frac{d_j}{D} \quad \text{(Pelton turbines)} \tag{8.4.11}$$

Practical values of d_j/D for Pelton wheels to ensure good efficiency are in the range 0.04 to 0.1, corresponding to N_s values in the range 0.03 to 0.08. Higher specific speeds are possible with multiple-nozzle designs. The increase is proportional to the square root of the number of nozzles. In considering an impulse unit, one must remember that efficiency is based on net head; the net head for an impulse unit is generally less than the net head for a reaction turbine at the same *gross head* because of the lack of a draft tube.

Reaction Turbines. The main difference between impulse and reaction turbines is the fact that a pressure drop takes place in the rotating passages of the reaction turbine. This implies that the entire flow passage from the turbine inlet to the discharge at the tailwater must be completely filled. A major factor in the overall design of modern reaction turbines is the draft tube. It is usually desirable to reduce the overall equipment and civil construction costs by using high-specific speed runners. Under these circumstances the draft tube is extremely critical both flow-stability and efficiency viewpoints.* At higher specific speed, a substantial percentage of the available total energy is in the form of kinetic energy leaving the runner. To recover this efficiently, considerable emphasis should be placed on the draft tube design.

The practical specific speed range for reaction turbines is much broader than for impulse wheels. This is due to the wider range of variables which control the basic operation of the turbine. The pivoted guide vanes allow for control of the magnitude and direction of the inlet flow. Because there is a fixed relationship among blade angle, inlet velocity, and peripheral speed for shock-free entry, this requirement cannot be completely satisfied at partial flow without the ability to vary blade angle. This is the distinction between the efficiency of fixed-propeller and Francis-types at partial loads and the fully adjustable Kaplan design.

Referring to Equation 8.4.4, optimum hydraulic efficiency of the runner would occur when α_2 is equal to 90°. However, the overall efficiency of the turbine is dependent on the optimum performance of the draft tube as well, which occurs with a little swirl in the flow. Thus, the best overall efficiency occurs with $\alpha_2 \approx 75°$ for high-specific speed turbines.

The determination of optimum specific speed in a reaction turbine is more complicated than for an impulse unit since there are more variables. For a radial-flow machine, an approximate expression is

$$N_s = 1.64\left[C_v \sin\alpha_1 \frac{B}{D_1}\right]^{1/2} \omega_{11} \quad \text{(Francis turbines)} \tag{8.4.12}$$

where C_v is the fraction of net head that is in the form of inlet velocity head and B is the height of the inlet flow passage (Figure 8.4.3). Value of N_s for Francis units is normally found to be in the range 0.3 to 2.5.

Standardized axial-flow machines are available in the smaller sizes. These units are made up of standard components, such as shafts and blades. For such cases,

$$N_s \sim \frac{\sqrt{\tan\beta}}{n_B^{3/4}} \quad \text{(propeller turbines)} \tag{8.4.13}$$

* This should be kept in mind when retrofitting an older, low-specific-speed turbine with a new runner of higher capacity.

where β is the blade pitch angle and n_B is the number of blades. The advantage of controllable pitch is also obvious from this formula, the best specific speed simply being a function of pitch angle.

It should be further noted that ω_{11} is approximately constant for Francis units and N_s is proportional to $(B/D_1)^{1/2}$. It can be also shown that velocity component based on the peripheral speed at the throat, ω_{11e}, is proportional to N_s. In the case of axial-flow machinery, ω_{11} is also proportional to N_s. For minimum cost, peripheral speed should be as high as possible, consistent with cavitation-free performance. Under these circumstances, N_s would vary inversely with the square root of head:

$$N_s = \frac{C}{\sqrt{H}} \quad (H \text{ is in meters}) \tag{8.4.14}$$

where the range of C is 21 to 30 for fixed propeller units, 21 to 32 for Kaplan units, and 16 to 25 for Francis units.

Performance Comparison. The physical characteristics of various runner configurations are summarized in Figure 8.4.8. It is obvious that the configuration changes with speed and head. Impulse turbines are efficient over a relatively narrow range of specific speed, whereas Francis and propeller turbines have a wider useful range. An important consideration is whether or not a turbine is required to operate over a wide range of load. Pelton wheels tend to operate efficiently over a wide range of power loading because of their nozzle design. In the case of reaction machines that have fixed geometry, such as Francis and propeller turbines, efficiency can vary widely with load. However, Kaplan and Deriaz* turbines can maintain high efficiency over a wide range of operating conditions. The decision of whether to select a simple configuration with a relatively "peaky" efficiency curve or go to the added expense of installing a more complex machine with a broad efficiency curve will depend on the expected operation of the plant and other economic factors.

Note in Figure 8.4.8 that there is an overlap in the range of application of various types of equipment. This means that either type of unit can be designed for good efficiency in this range, but other factors, such as generator speed and cavitation, may dictate the final selection.

Speed Regulation

The speed regulation of a turbine is an important and complicated problem. The magnitude of the problem varies with size, type of machine and installation, type of electrical load, and whether or not the plant is tied into an electrical grid. Note that runaway or no-load speed can be higher than design speed by factors as high as 2.6. This is an important design consideration for all rotating parts, including the generator.

The speed of a turbine has to be controlled to a value that matches the generator characteristics and the grid frequency:

$$n = \frac{120f}{N_p} \tag{8.4.15}$$

where n is turbine speed in rpm, f is the required grid frequency in Hz, and N_p is the number of poles in the generator. Typically, N_p is in multiples of 4. There is a tendency to select higher speed generators to minimize weight and cost. However, consideration has to be given to speed regulation.

It is beyond the scope of this section to discuss the question of speed regulation in detail. Regulation of speed is normally accomplished through flow control. Adequate control requires sufficient rotational inertia of the rotating parts. When load is rejected, power is absorbed, accelerating the flywheel; and when load is applied, some additional power is available from deceleration of the flywheel. Response

* An adjustable blade mixed-flow turbine (Arndt, 1991).

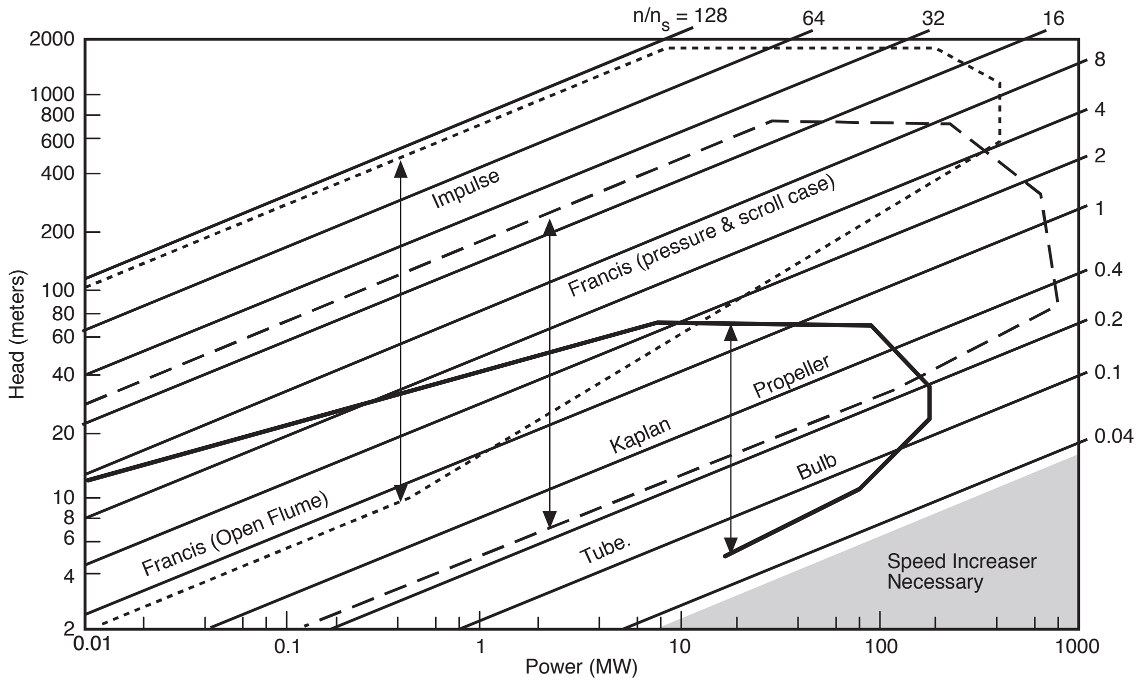

FIGURE 8.4.8 Application chart for various turbine types (n/n_S is the ratio of turbine speed in rpm, n, to specific speed defined in the metric system, $n_S = nP^{1/2}/H^{3/4}$ with P in kilowatts). (From Arndt, R.E.A., in *Hydropower Engineering Handbook*, J.S. Gulliver and R.E.A. Arndt, Eds., McGraw-Hill, New York, 1991, 4.1–4.67. With permission.)

time of the governor must be carefully selected, since rapid closing can lead to excessive pressures in the penstock.

A Francis turbine is controlled by opening and closing the wicket gates, which vary the flow of water according to the load. The actuator components of a governor are required to overcome the hydraulic and frictional forces and to maintain the wicket gates in fixed position under steady load. For this reason, most governors have hydraulic actuators. On the other hand, impulse turbines are more easily controlled, because the jet can be deflected or an auxiliary jet can by bypass flow from the power-producing jet without changing the flow rate in the penstock. This permits long delay times for adjusting the flow rate to the new flywheel; and when load is applied power conditions. The spear or needle valve controlling the flow rate can close quite slowly, say, in 30 to 60 sec, thereby minimizing pressure rise in the penstock.

Several types of governors are available, which vary with the work capacity desired and/or the degree of sophistication of control. These vary from pure mechanical to mechanical-hydraulic and electrohydraulic. Electrohydraulic units are sophisticated pieces of equipment and would not be suitable for remote regions. The precision of governing necessary will depend on whether the electrical generator is synchronous or asynchronous (induction type). There are advantages to the induction type of generator. It is less complex and therefore less expensive, but has typically slightly lower efficiency. Its frequency is controlled by the frequency of the grid it is feeding into, thereby eliminating need of an expensive the conventional governor. It cannot operate independently but can only feed into a network and does so with lagging power factor which may or may not be a disadvantage, depending on the nature of the load. Long transmission lines, for example, have a high capacitance and in this case the lagging power factor may be an advantage.

Speed regulation is a function of the flywheel effect of the rotating components and the inertia of the water column of the system. The start-up time of the rotating system is given by

$$t_s = \frac{I\omega^2}{P} \tag{8.4.16}$$

where I = moment of inertia of the generator and turbine, kg · m^2 (Bureau of Reclamation, 1966).

The start-up time of the water column is given by

$$t_p = \frac{\sum LV}{gH} \tag{8.4.17}$$

where L = length of water column
V = velocity in each component of the water column

For good speed regulation, it is desirable to keep $t_s/t_p > 4$. Lower values can also be used, although special precautions are necessary in the control equipment. Higher ratios of t_s/t_p can be obtained by increasing I or decreasing t_p. Increasing I implies a larger generator, which also results in higher costs. The start-up time of the water column can be reduced by reducing the length of the flow system, by using lower velocities, or by adding **surge tanks**, which essentially reduce the effective length of the conduit. A detailed analysis should be made for each installation, since for a given length, head, and discharge the flow area must be increased to reduce t_p, which leads to associated higher construction costs.

Cavitation and Turbine Setting

Another factor that must be considered prior to equipment selection is the evaluation of the turbine with respect to tailwater elevations. Hydraulic turbines are subject to pitting due to cavitation (Arndt, 1981, 1991). For a given head, smaller, lower-cost, high-speed runner must be set lower (i.e, closer to tailwater or even below tailwater) than a larger, higher-cost, low-speed turbine runner. Also, atmospheric pressure or elevation above sea level is a factor, as are tailwater elevation ion variations and operating requirements. This is a complex concept which can only be accurately resolved by model tests. The runner design will have different cavitation characteristics. Therefore, the anticipated turbine location or setting with respect to tailwater elevations is an important consideration in turbine selection.

Cavitation is not normally a problem with impulse wheels. However, by the very nature of their operation, cavitation is an important factor in reaction turbine installations. The susceptibility for cavitation to occur is a function of the installation and the turbine design. This can be expressed conveniently in terms of Thoma's sigma defined as

$$\sigma_T = \frac{H_a - H_v - z}{H} \tag{8.4.18}$$

where H_a is the atmospheric pressure head, H_V is the vapor pressure need (generally negligible), and z is the elevation of a turbine reference plane above the tailwater (see Figure 8.4.1). Draft tube losses and the exit velocity head have been neglected.

σ_T must be above a certain value to avoid cavitation problems. The critical value of σ_T is a function of specific speed (Arndt, 1991). The Bureau of Reclamation (1966) suggests that cavitation problems can be avoided when

$$\sigma_T > 0.26 N_s^{1.64} \tag{8.4.19}$$

Equation 8.4.19 does not guarantee total elimination of cavitation, only that cavitation is within acceptable limits. Cavitation can be totally avoided only if the value σ_T at an installation is much greater than the limiting value given in Equation 8.4.19. The value of σ_T for a given installation is known as the plant sigmas, σ_p. Equation 8.4.19 should only be considered as a guide in selecting σ_p, which is normally determined by a model test in the manufacturer's laboratory. For a turbine operating under a given head, the only variable controlling σ_p is the turbine setting z. The required value of σ_p then controls the allowable setting above tailwater:

$$z_{\text{allow}} = H_a - H_v - \sigma_p H \tag{8.4.20}$$

It must be borne in mind that H_a varies with elevation. As a rule of thumb, H_a decreases from the sea-level value of 10.3 m by 1.1 m for every 1000 m above sea level.

Defining Terms

Draft tube: The outlet conduit from a turbine which normally acts as a diffuser. This is normally considered to be an integral part of the unit.

Forebay: The hydraulic structure used to withdraw water from a reservoir or river. This can be positioned a considerable distance upstream from the turbine inlet.

Head: The specific energy per unit weight of water. *Gross head* is the difference in water surface elevation between the forebay and tailrace. *Net head* is the difference between *total head* (the sum of velocity head $V^2/2g$, pressure head $p/\rho g$, and elevation head z at the inlet and outlet of a turbine. Some European texts use specific energy per unit mass, e.g., specific kinetic energy is $V^2/2$.

Runner: The rotating component of a turbine in which energy conversion takes place.

Specific speed: A universal number for a given machine design.

Spiral case: The inlet to a reaction turbine.

Surge tank: A hydraulic structure used to diminish overpressures in high-head facilities due to water hammer resulting from the sudden stoppage of a turbine

Wicket gates: Pivoted, streamlined guide vanes that control the flow of water to the turbine.

References

Arndt, R.E.A. 1981. Cavitation in fluid machinery and hydraulic structures. *Ann. Rev. Fluid Mech.* 13:273–328.

Arndt, R.E.A. 1991. Hydraulic turbines, in *Hydropower Engineering Handbook,* J.S. Gulliver and R.E.A. Arndt, Eds., pp. 4.1–4.67 McGraw-Hill, New York.

Bureau of Reclamation. 1966. Selecting Hydraulic Reaction Turbines, Engineering Monograph No. 20.

Daily, J.W. 1950. Hydraulic machinery, in H. Rouse, Ed., Engineering Hydraulics, New York.

International Code for the Field Acceptance Tests of Hydraulic Turbines, 1963. International Electrotechnical Commission, Publication 41.

Raabe, J. 1985. *Hydro Power: The Design, Use, and Function of Hydromechanical, Hydraulic, and Electrical Equipment*. VDI Verlag, Dusseldorf, Germany.

Wahl, T.L. 1994. Draft tube surging times two: The twin vortex problem. *Hydro Rev.* 13(1):60–69.

Further Information

J. Fluids Eng, published quarterly by the ASME.

ASME Symposia Proc. on Fluid Machinery and Cavitation, published by the Fluids Eng. Div.

Hydro Review, published eight times per year by HCI Publications, Inc., Kansas City, MO.

L.F. Moody and T. Zowski, Hydraulic machinery, in *Handbook of Applied Hydraulics,* C.V. Davis and K.E. Sorenson, Eds., McGraw-Hill, New York, 1992.

Waterpower and Dam Construction, published monthly by Reed Business Publishing, Surrey, U.K.

8.5 Stirling Engines

William B. Stine

Introduction

The Stirling engine was patented in 1816 by Rev. Robert Stirling, a Scottish minister (Figure 8.5.1). Early Stirling engines were coal-burning, low-pressure air engines built to compete with saturated steam engines for providing auxiliary power for manufacturing and mining. In 1987, John Ericsson built an enormous marine Stirling engine with four 4.2-m-diameter pistons. Beginning in the 1930s, the Stirling engine was brought to a high state of technology development by Philips Research Laboratory in Eindhoven, The Netherlands with the goal of producing small, quiet electrical generator sets to be used with high-power-consuming vacuum tube electronic devices. Recently, interest in Stirling engines has resurfaced, with solar electric power generation (Stine and Diver, 1994) and hybrid automotive applications in the forefront.

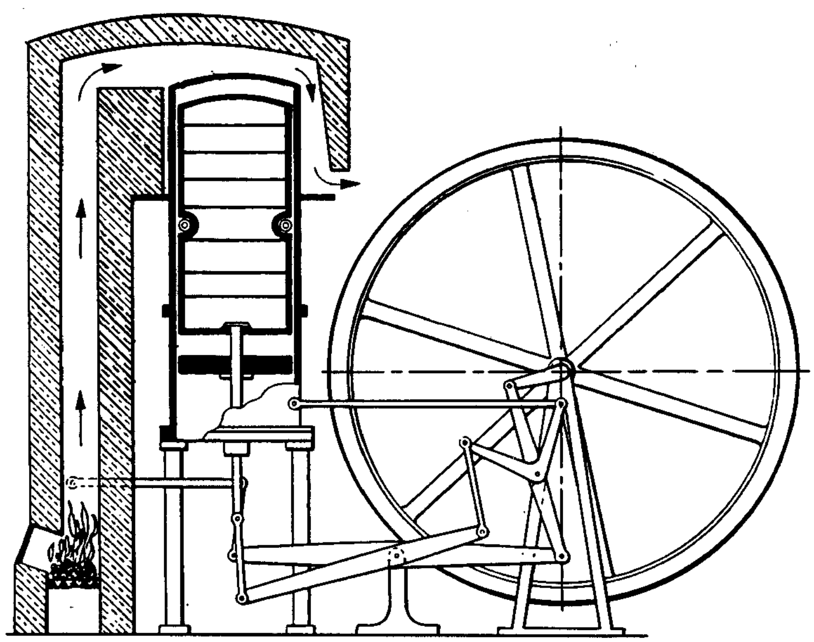

FIGURE 8.5.1 The original patent Stirling engine of Rev. Robert Stirling.

In theory, the **Stirling cycle** engine can be the most efficient device for converting heat into mechanical work with high efficiencies requiring high-temperatures. In fact, with regeneration, the efficiency of the Stirling cycle equals that of the Carnot cycle, the most efficient of all ideal thermodynamic cycles. (See West, 1986 for further discussion of the thermodynamics of Stirling cycle machines.)

Since their invention, prototype Stirling engines have been developed for automotive purposes; they have also been designed and tested for service in trucks, buses, and boats (Walker, 1973). The Stirling engine has been proposed as a propulsion engine in yachts, passenger ships, and road vehicles such as city buses (Meijer, 1992). The Stirling engine has also been developed as an underwater power unit for submarines, and the feasibility of using the Stirling for high-power space-borne systems has been explored by NASA (West, 1986). The Stirling engine is considered ideal for solar heating, and the first solar application of record was by John Ericsson, the famous British-American inventor, in 1872 (Stine and Diver, 1994).

Stirling engines are generally externally heated engines. Therefore, most sources of heat can be used to drive them, including combustion of just about anything, radioisotopes, solar energy, and exothermic chemical reactions. High-performance Stirling engines operate at the thermal limits of the materials used for their construction. Typical temperatures range fiom 650 to 800°C (1200 to 1470°F), resulting in engine conversion efficiencies of around 30 to 40%. Engine speeds of 2000 to 4000 rpm are common

Thermodynamic Implementation of the Stirling Cycle

In the ideal Stirling cycle, a **working gas** is alternately heated and cooled as it is compressed and expanded. Gases such as helium and hydrogen, which permit rapid heat transfer and do not change phase, are typically used in the high-performance Stirling engines. The ideal Stirling cycle combines four processes, two constant-temperature heat-exchange processes and two constant-volume heat-exchange processes. Because more work is done by expanding high-temperature, high-pressure gas than is required to compress low-temperature, low-pressure gas, the Stirling cycle produces net work, which can drive an electric alternator or other mechanical devices.

Working Gases

In the Stirling cycle, the working gas is alternately heated and cooled in constant-temperature and constant-volume processes. The traditional gas for Stirling engines has been air at atmospheric pressure. At this pressure, air has a reasonably high density and therefore can be used directly in the cycle with loss of working gas through seals a minor problem. However, internal component temperatures are limited because of the oxygen in air which can degrade materials rapidly.

Because of their high heat-transfer capabilities, hydrogen and helium are used for high-speed, high-performance Stirling engines. To compensate for the low density of these gases, the mean pressure of the working gas is raised by charging the gas spaces in the engine to high pressures. Compression and expansion vary above and below this **charge pressure**. Hydrogen, thermodynamically a better choice, generally results in more-efficient engines than does helium (Walker, 1980). Helium, on the other hand, has fewer material-compatibility problems and is safer to work with. To maximize power, high-performance engines typically operate at high pressure, in the range of 5 to 20 MPa (725 to 2900 psi). Operation at these high gas pressures makes sealing difficult, and seals between the high-pressure region of the engine and those parts at ambient pressure have been problematic in some engines. New designs to reduce or eliminate this problem are currently being developed.

Heat Exchange

The working gas is heated and cooled by heat exchangers that add heat from an external source, or reject heat to the surroundings. Further, in most engines, an internal heat storage unit stores and rejects heat during each cycle.

The **heater** of a Stirling engine is usually made of many small-bore tubes that are heated externally with the working gas passing through the inside. External heat transfer by direct impingement of combustion products or direct adsorption of solar irradiation is common. A trade-off between high heat-transfer rate using many small-bore tubes with resulting pumping losses, and fewer large-bore tubes and lower pumping losses drives the design. A third criterion is that the volume of gas trapped within these heat exchangers should be minimal to enhance engine performance. In an attempt to provide more uniform and constant-temperature heat transfer to the heater tubes, **reflux** heaters are being developed (Stine and Diver, 1994). Typically, by using sodium as the heat-transfer medium, liquid is evaporated at the heat source and is then condensed on the outside surfaces of the engine heater tubes.

The **cooler** is usually a tube-and-shell heat exchanger. Working gas is passed through the tubes, and cooling water is circulated around the outside. The cooling water is then cooled in an external heat exchanger. Because all of the heat rejected from the power cycle comes from the cooler, the Stirling engine is considered ideal for cogeneration applications.

Most Stirling engines incorporate an efficiency-enhancing **regenerator** that captures heat from the working gas during constant-volume cooling and replaces it when the gas is heated at constant volume. Heating and cooling of the regenerator occurs at over 60 times a second during high-speed engine operation. In the ideal cycle, all of the heat-transferred during the constant volume heating and cooling processes occurs in the regenerator, permitting the external heat addition and rejection to be efficient constant-temperature heat-transfer processes. Regenerators are typically chambers packed with fine-mesh screen wire or porous metal structures. There is enough thermal mass in the packing material to store all of the heat necessary to raise the temperature of the working gas from its low to its high temperature. The amount of heat stored by the regenerator is generally many times greater than the amount added by the heater.

Power Control

Rapid control of the output power of a Stirling engine is highly desirable for some applications such as automotive and solar electric applications. In most Stirling engine designs, rapid power control is implemented by varying the density (i.e., the mean pressure) of the working gas by bleeding gas from the cycle when less power is desired. To return to a higher power level, high-pressure gas must be reintroduced into the cycle. To accomplish this quickly and without loss of working gas, a complex system of valves, a temporary storage tank, and a compressor are used.

A novel method of controlling the power output is to change the length of stroke of the power piston. This can be accomplished using a variable-angle swash plate drive as described below. Stirling Thermal Motors, Inc., uses this method on their STM 4-120 Stirling engine (Figure 8.5.2).

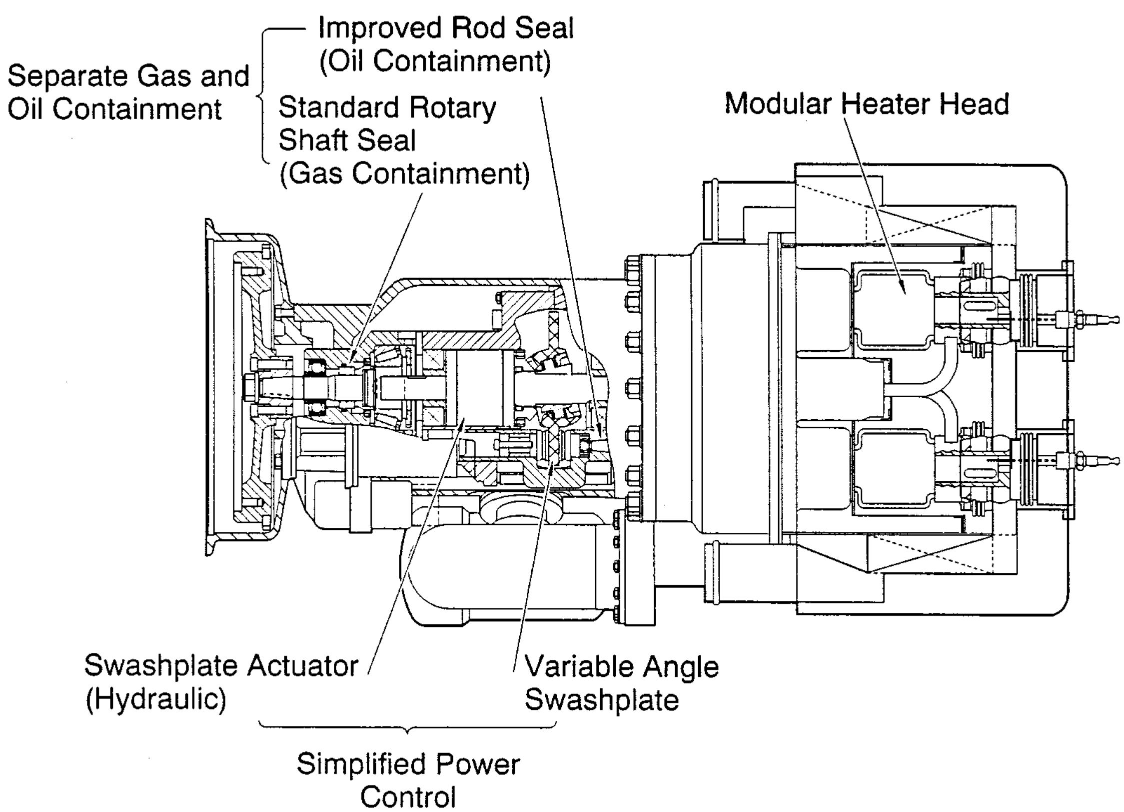

FIGURE 8.5.2 Stirling Thermal Motors 4-120 variable swash plate Rinia configuration engine. (Courtesy Stirling Thermal Motors, Ann Arbor, Michigan.)

Mechanical Implementation of the Stirling Cycle

Piston/Displacer Configurations

To implement the Stirling cycle, different combinations of machine components have been designed to provide for the constant-volume movement of the working gas between the high- and low-temperature regions of the engine, and compression and expansion during the constant-temperature heating and cooling. The compression and expansion part of the cycle generally take place in a cylinder with a piston. Movement of the working gas back and forth through the heater, regenerator, and cooler at constant volume is often implemented by a **displacer**. A displacer in this sense is a hollow plug that, when moved to the cold region, displaces the working gas from the cold region causing it to flow to the hot region and vice versa. Only a small pressure difference exists between either end of the displacer, and, therefore, sealing requirements and the force required to move it are minimal.

Three different design configurations are generally used (Figure 8.5.3). Called the alpha-, beta-, and gamma-configurations. Each has its distinct mechanical design characteristics, but the thermodynamic cycle is the same. The **alpha-configuration** uses two pistons on either side of the heater, regenerator, and the cooler. These pistons first move uniformly in the same direction to provide constant-volume processes to heat or cool the gas. When all of the gas has been moved into one cylinder, one piston remains fixed and the other moves to compress or expand the gas. Compression work is done by the **cold piston** and expansion work done on the **hot piston** The alpha-configuration does not use a displacer. The Stirling Power Systems V-160 engine (Figure 8.5.4) is an example of this configuration.

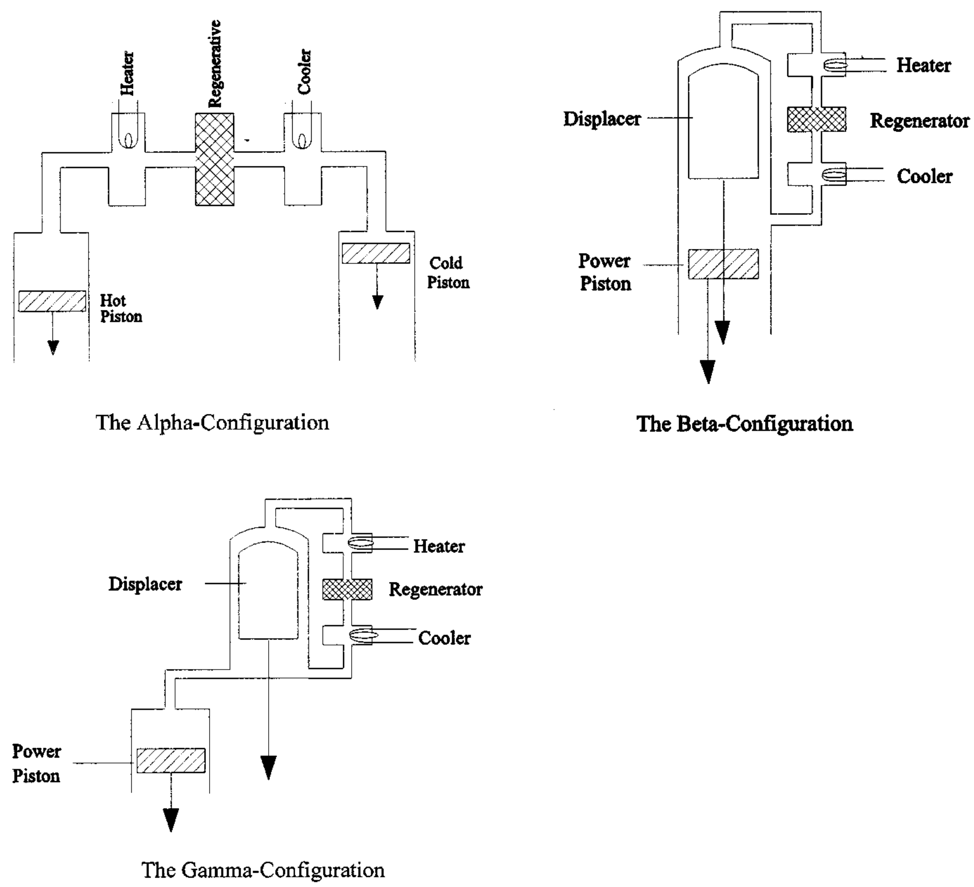

FIGURE 8.5.3 Three fundamental mechanical configurations for Stirling Engines.

A variation on using two separate pistons to implement the alpha-configuration is to use the front and back side of a single piston called a **double-acting piston**. The volume at the front side of one piston is connected, through the heater, regenerator, and cooler, to the volume at the back side of another piston. With four such double-acting pistons, each 90° out of phase with the next, the result is a four-cylinder alpha-configuration engine. This design is called the *Rinia* or *Siemens configuration* and the United Stirling 4-95 (Figure 8.5.5) and the Stirling Thermal Motors STM 4-120 (Figure 8.5.2) are current examples.

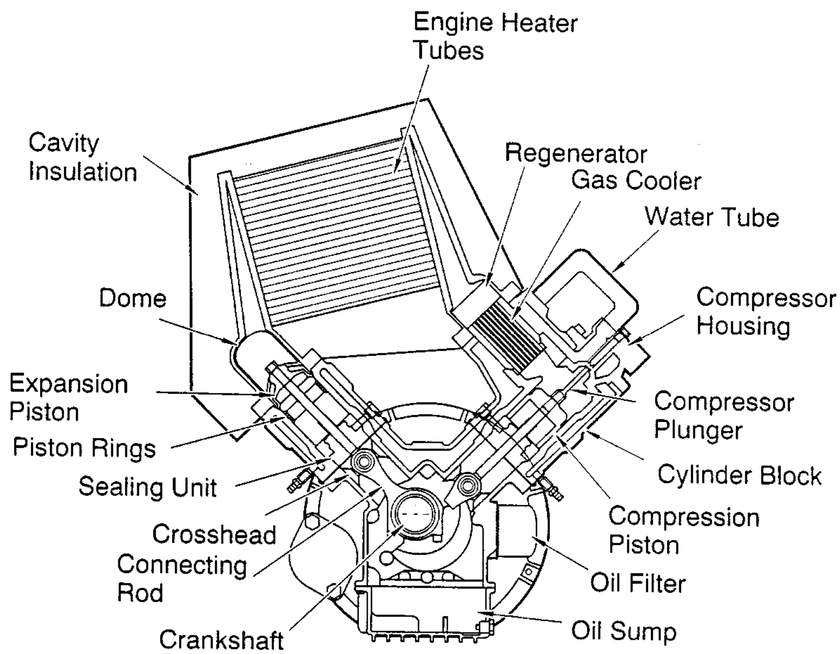

FIGURE 8.5.4 Stirling Power Systems/Solo Kleinmotoren V-160 alpha-configuration Stirling engine.

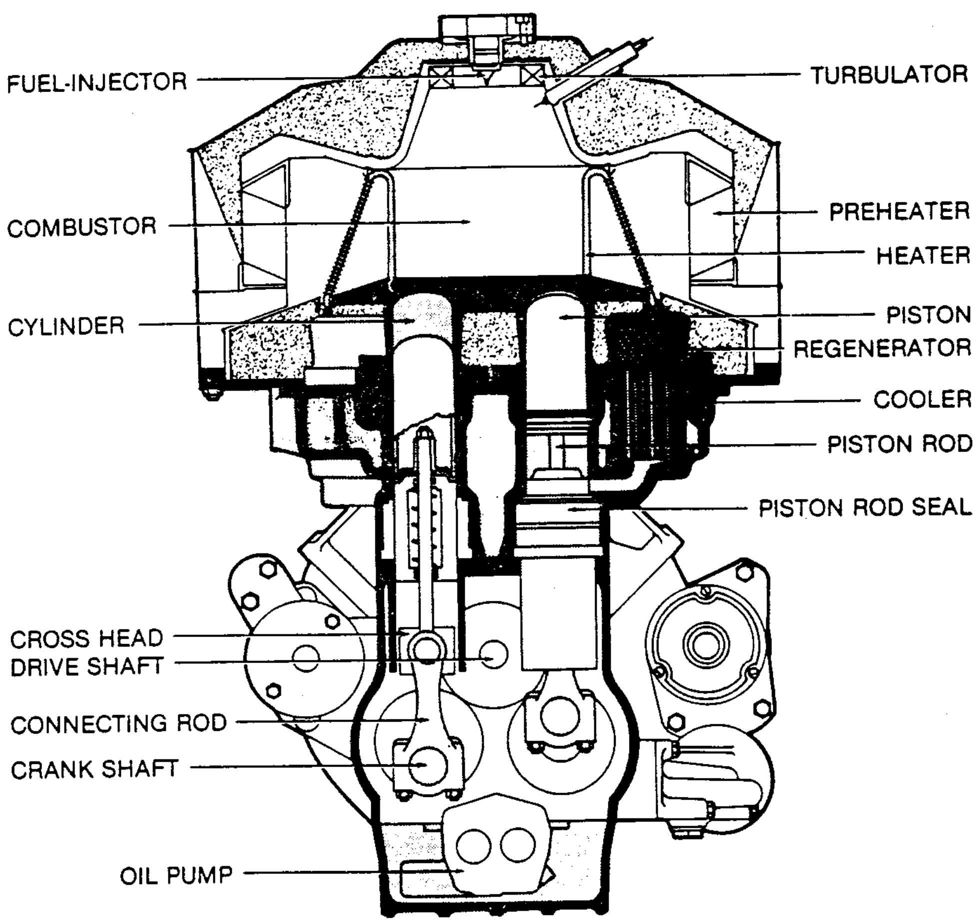

FIGURE 8.5.5 The 4-95 high-performance Siemens configuration Rinia engine by United Stirling (Malmo, Sweden).

The **beta-configuration** is a design incorporating a displacer and a power piston in the same cylinder. The displacer shuttles gas between the hot end and the cold end of the cylinder through the heater, regenerator, and cooler. The power piston, usually located at the cool end of the cylinder, compresses the working gas when the gas is in the cool end and expands the working gas when the gas has been moved to the hot end. The original patent engine by Robert Stirling and most free-piston Stirling engines discussed below are of the beta-configuration.

The third configuration, using separate cylinders for the displacer and the power piston, is called the **gamma-configuration**. Here, the displacer shuttles gas between the hot end and the cold end of a cylinder through the heater, regenerator, and cooler, just as with the beta-configuration. However, the power piston is in a separate cylinder, pneumatically connected to the displacer cylinder.

Piston/Displacer Drives

Most Stirling engine designs dynamically approximate the Stirling cycle by moving the piston and displacer with **simple harmonic motion**, either through a crankshaft, or bouncing as a spring/mass second-order mechanical system. For both, a performance penalty comes from the inability of simple harmonic motion to perfectly follow the desired thermodynamic processes. A more desirable dynamic from the cycle point of view, called overdriven or **bang-bang motion**, has been implemented in some designs, most notably the **Ringbom configuration** and engines designed by Ivo Kolin (West, 1986).

Kinematic Engines. Stirling engine designs are usually categorized as either kinematic or free-piston. The power piston of a **kinematic Stirling engine** is mechanically connected to a rotating output shaft. In typical configurations, the power piston is connected to the crankshaft with a connecting rod. In order to eliminate side forces against the cylinder wall, a **cross-head** is often incorporated, where the connecting rod connects to the cross-head, which is laterally restrained so that it can only move linearly in the same direction as the piston. The power piston is connected to the cross-head and therefore experiences no lateral forces. The critical sealing between the high-pressure and low-pressure regions of the engine can now be created using a simple **linear seal** on the shaft between the cross-head and the piston. This design also keeps lubricated bearing surfaces in the low-pressure region of the engine, reducing the possibility of fouling heat-exchange surfaces in the high-pressure region of the engine. If there is a separate displacer piston as in the beta- and gamma configurations, it is also mechanically connected to the output shaft.

A variation on crankshaft/cross-head drives is the **swash plate** or **wobble-plate drive**, used with success in some Stirling engine designs. Here, a drive surface affixed to the drive shaft at an angle, pushes fixed piston **push rods** up and down as the slanted surface rotates beneath. The length of stroke for the piston depends on the angle of the plate relative to the axis of rotation. The STM 4-120 engine (Figure 8.5.2) currently being commercialized by Stirling Thermal Motors incorporates a **variable-angle swash plate drive** that permits variation in the length of stroke of the pistons.

Free-Piston Engine/Converters. An innovative way of accomplishing the Stirling cycle is employed in the free-piston engine. In this configuration, the power piston is not mechanically connected to an output shaft. It bounces alternately between the space containing the working gas and a spring (usually a gas spring). In many designs, the displacer is also free to bounce on **gas springs** or mechanical springs (Figure 8.5.6). This configuration is called the Beale free-piston Stirling engine after its inventor, William Beale. Piston stroke, frequency, and the timing between the two pistons are established by the dynamics of the spring/mass system coupled with the variations in cycle pressure. To extract power, a magnet can be attached to the power piston and electric power generated as it moves past stationary coils. These Stirling engine/alternator units are called **free-piston Stirling converters**. Other schemes for extracting power from free-piston engines, such as driving a hydraulic pump, have also been considered.

Free-piston Stirling engines have only two moving parts, and therefore the potential advantages of simplicity, low cost, and ultra-reliability. Because electricity is generated internally, there are no dynamic seals between the high-pressure region of the engine and ambient, and no oil lubrication is required. This design promises long lifetimes with minimal maintenance. A number of companies are currently

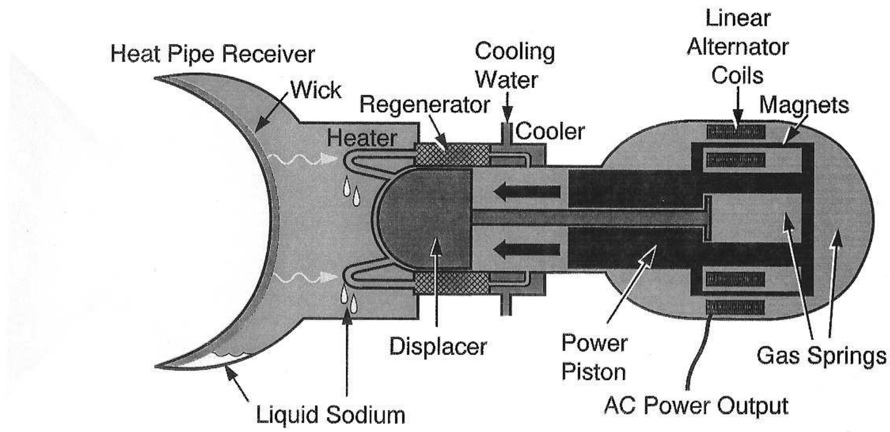

FIGURE 8.5.6 Basic components of a Beale free-piston Stirling converter incorporating a sodium heat pipe receiver for heating with concentrated solar energy.

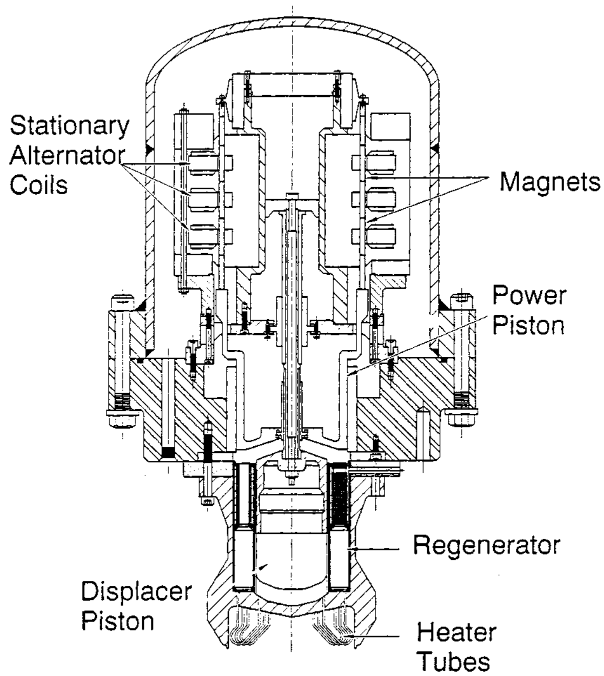

FIGURE 8.5.7 The Sunpower 9-kWe free-piston beta-configuration Stirling engine.

developing free-piston engines including Sunpower, Inc. (Figure 8.5.7), and Stirling Technology Company.

Seals and Bearings

Many proposed applications for Stirling engine systems require long-life designs. To make systems economical, a system lifetime of at least 20 years with minimum maintenance is generally required. Desired engine lifetimes for electric power production are 40,000 to 60,000 hr — approximately ten times longer than that of a typical automotive internal combustion engine. Major overhaul of engines, including replacement of seals and bearings, may be necessary within the 40,000- to 60,000-hr lifetime, which adds to the operating cost. A major challenge, therefore, in the design of Stirling engines is to reduce the potential for wear in critical components or to create novel ways for them to perform their tasks.

Piston seals differ from those used in internal combustion engines in a number of ways. Sealing of the power piston is critical since **blow-by loss** of the hydrogen or helium working gas must be captured and recompressed, or replaced from a high-pressure cylinder. Displacer sealing is less critical and only necessary to force most of the working gas through the heater, regenerator, and cooler. Oil for friction reduction or sealing cannot be used because of the danger of it getting into the working gas and fouling the heat-exchange surfaces. This leads to two choices for sealing of pistons, using **polymer sealing rings** or **gas bearings** (simply close tolerance fitting between piston and wall).

Free-piston engines with gas springs have special internal sealing problems. Small leakage can change the force-position characteristics of the "spring" and rapidly upset the phase and displacement dynamics designed into the engine. In order to prevent this, *centering ports* are used to ensure that the piston stays at a certain location; however, these represent a loss of potential work from the engine.

Materials

Materials used in Stirling engines are generally normal steels with a few exceptions. Materials that can withstand continuous operation at the cycle high temperature are required for the heater, regenerator, and the hot side of the displacement volume. Because most engines operate at high pressure, thick walls are often required. In the hot regions of the engine, this can lead to *thermal creep* due to successive heating and cooling. In the cool regions, large spaces for mechanical linkages can require heavy, thick walls to contain the gas pressure. Use of composite structure technology or reducing the size of the pressurized space can eliminate these problems.

Future of the Stirling Engine

The principal advantages of the Stirling engine, external heating and high efficiency, make this the engine of the future, replacing many applications currently using internal combustion engines and providing access to the sun as an inexpensive source of energy (Figure 8.5.6). For hybrid-electric automotive applications, the Stirling engine is not only almost twice as efficient as modern spark-ignition engines, but because of the continuous combustion process, it burns fuel more cleanly and is not sensitive to the quality or type of fuel. Because of the simplicity of its design, the Stirling engine can be manufactured as an inexpensive power source for electricity generation using biomass and other fuels available in developing nations.

Most importantly, the Stirling engine will provide access to inexpensive solar energy. Because it can receive its heat from a resource 93 million miles away using concentrating solar collectors, and because its manufacture is quite similar to the gasoline or diesel engine, and because economies of scale are certain when producing thousands of units per year, the Stirling engine is considered to be the least expensive alternative for solar energy electric generation applications in the range from 1 kWe to 100 MWe.

Defining Terms

Alpha-configuration: Design of a Stirling engine where two pistons moving out of phase, and cause the working gas between them to go through the four processes of the Stirling cycle.

Beale free-piston Stirling engine: Stirling engine configuration where the power piston and displacer in a single cylinder are free to bounce back and forth along a single axis, causing the enclosed working gas to go through the four processes of the Stirling cycle. Restoration forces are provided by the varying pressure of the working gas, springs (gas or mechanical), and the external load which can be a linear alternator or a fluid pump.

Beta-configuration: Design of a Stirling engine where the displacer and power piston are located in the same cylinder and cause the enclosed working gas to go through the four processes of the Stirling cycle.

Blow-by: The gas that leaks past a seal, especially a piston-to-cylinder seal.

Charge pressure: Initial pressure of the working gas.

Cooler: Heat exchanger that removes heat from the working fluid and transfers it out of the cycle.

Cross-head: A linear sliding bearing surface connected to a crankshaft by a connecting rod. It is purpose is to provide linear reciprocating motion along a single line of action.

Displacer: Closed volume 'plug' that forces the working fluid to move from one region of the engine to another by displacing it.

Double-acting piston: A piston in an enclosed cylinder where pressure can be varied on both sides of the piston, resulting a total amount of work being the sum of that done on or by both sides.

Dynamic seals: Seals that permit transfer of motion without permitting gas or oil leakage. These can be either *linear seals* permitting a shaft to move between two regions (i.e., the piston rod seals in a Stirling engine), or *rotating seals* that permit rotating motion to be transmitted from one region to another (i.e., the output shaft of a Stirling engine).

Free-piston Stirling converters: A name given to a hermetically sealed free-piston Stirling engine incorporating an internal alternator or pump.

Gamma-configuration: A design of a Stirling engine where the displacer and power piston are located in separate, connected cylinders and cause the enclosed working gas to go through the four processes of the Stirling cycle.

Gas bearing: A method of implementing the sliding seal between a piston and cylinder as opposed to using piston rings. Uses a precision-fitting piston that depends on the small clearance and long path for sealing and on the viscosity of the gas for lubrication.

Gas spring: A piston that compresses gas in a closed cylinder where the restoration force is linearly proportional to the piston displacement. This is a concept used in the design of free-piston Stirling engines.

Heater: A heat exchanger which adds heat to the working fluid from an external source.

Kinematic stirling engine: Stirling engine design that employ physical connections between the power piston, displacer, and a mechanical output shaft.

Linear seal: see **dynamic seals**.

Overdriven (bang-bang) motion: Linear motion varying with time as a square-wave function. An alternative to simple harmonic motion and considered a better motion for the displacer of a Stirling engine but difficult to implement.

Phase angle: The angle difference between displacer and power piston harmonic motion with a complete cycle representing 360°. In most Stirling engines, the harmonic motion of the power piston follows (lags) the motion of the displacer by approximately 90°.

Push rod: A thin rod connected to the back of the piston that transfers linear motion through a dynamic linear seal, between the low- and high-pressure regions of an engine.

Reflux: A constant-temperature heat-exchange process where a liquid is evaporated by heat addition and then condensed as a result of cooling.

Regenerator: A heat-transfer device that stores heat from the working gas during part of a thermodynamic cycle and returns it during another part of the cycle. In the Stirling cycle the regenerator stores heat from one constant-volume process and returns it in the other constant-volume process.

Ringbom configuration: A Stirling engine configuration where the power piston is kinematically connected to a power shaft, and the displacer is a free piston that is powered by the difference in pressure between the internal gas and atmospheric pressure.

Simple harmonic motion: Linear motion varying with time as a sine function. Approximated by a piston connected to a crankshaft or a bouncing of a spring mass system.

Stirling cycle: A thermodynamic power cycle with two constant-volume heat addition and rejection processes and two constant-temperature heat-addition and rejection processes.

Swash plate drive: A disk on a shaft, where the plane of the disk is tilted away from the axis of rotation of the shaft. Piston push rods that move parallel to the axis of rotation but are displaced from the axis of rotation, slide on the surface of the rotating swash plate and therefore move up and down.

Variable-angle swash plate drive: A swash plate drive where the tilt angle between the drive shaft and the plate can be varied, resulting in a change in the displacement of the push rods.

Wobble plate drive: Another name for a swash plate drive.

Working gas: Gas within the engine that exhibits pressure and temperature change as it is heated or cooled and/or compressed or expanded.

References

Meijer, R.F. 1992. Stirling engine, in *McGraw-Hill Encyclopedia of Science and Technology*, 7th ed., pp. 440–445, McGraw-Hill, New York.

Stine, W.B. and Diver, R.E. 1994. A *Compendium of Solar Dish Stirling Technology,* Report SAND94-7026, Sandia National Laboratories, Albuquerque, NM 87185.

Walker, G. 1973. The Stirling engine, *Sci. Am.,* 229(2):80–87.

Walker, G. 1980. *Stirling Engines,* Clarendon Press, Oxford.

West, C.D. 1986. *Principles and Applications of Stirling Engines,* Van Nostrand Reinhold, New York.

Further Information

Books

Hargraves, C.M. *The Philips Stirling Engine,* Elsevier Press, London, 1991.

Organ, A.J. *Thermodynamics and Gas Dynamics of the Stirling Cycle Machine,* Cambridge University Press, Cambridge, 1992.

Senft, J.R. *Ringbom Stirling Engines,* Oxford University Press, Oxford, 1993.

Stine, W.B. and R.E. Diver, A *Compendium of Solar Dish/Stirling Technology,* SAND93-7026, Sandia National Laboratory, Albuquerque, 1994.

Urieli, I. and D.M. Berchowitz, *Stirling Cycle Engine Analysis,* Adam Hilger, Bristol, 1984

Walker, G. *Stirling Engines,* Clarendon Press, Oxford, 1980.

Walker, G. and J.R. Senft, *Free-Piston Stirling Engines,* Springer-Verlag, New York, 1985.

Walker, G., G. Reader, O.R. Fauvel, E.R. Bingham, *The Stirling Alternative,* Bordon & Breach, New York, 1994.

West, C.D. *Principles and Applications of Stirling Engines,* Van Nostrand Reinhold, New York, 1986.

Periodicals

Proceedings of the Intersociety Energy Conversion Engineering Conference (IECEC), published annually.

Stirling Machine World, a quarterly newsletter devoted to advancements in Stirling engines, edited by Brad Ross, 1823 Hummingbird Court, West Richland, WA 99353-9542.

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